ASIC Design Flow Tutorial

Lab 2b: Formal Verification and Post-synthesis Simulation

Note: source /home/share/Env/synopsys_setup.env to open all tools.

A. Laboratory tasks

In Formality the following concepts are used:

Reference design: This design is the golden design, the standard against which Formality tests for equivalence.

Implementation design: This design is the changed design. It is the design whose accuracy you want to prove. For example, a newly synthesized design is an implementation of the source RTL design.

To start Formality, enter the following command at the terminal:

```
% fm_shell
fm_shell (setup)>
```

The fm_shell command starts the Formality shell environment. From here, start the graphical user interface (GUI) as follows:

```
fm_shell (setup)> start_gui
```

When Formality is invoked, begin in the **setup** mode. The **setup** indicates the mode that currently in when using commands. The modes are **setup**, **match**, and **verify**.

This laboratory work includes the following sections:
0. Guidance (Load Automated Setup File)
1. Reference (Specify the Reference Design)
2. Implementation (Specify the Implementation Design)
3. Set up (Set Up the Design)
4. Match (Match Compare Points)
5. Verify (Verify the Designs)
6. Debug.

0. Guidance (Load Automated Setup File)

Before specifying the reference and implementation designs, an automated setup file (.svf) can be optionally loaded into Formality. The automated setup file helps Formality process design changes caused by other tools used in the design flow. Formality uses this file to assist the compare point matching and verification process. For each automated setup file that is loaded, Formality processes the content and stores the information for use during the name-based compare point matching period.

If Formality is given the result of DC, then work in the **~/TH_TKVM_CLC/Digital_Synopsys/Formal/Post_syn** directory, and if Formality is given the result of ICC2, then work in the **~/TH_TKVM_CLC/Digital_Synopsys/Formal/Post_PnR** directory. This lab shows an example where Formality works with the result of DC. To run Formality, which works with the result of ICC2 is similar to this lab steps.

Start Formality graphical user interface (GUI) from work directory, which is located in **Post_syn** directory. To start it, enter:

```
% fm_shell -gui
```

This opens the Formality top-level GUI window (Fig. 1).

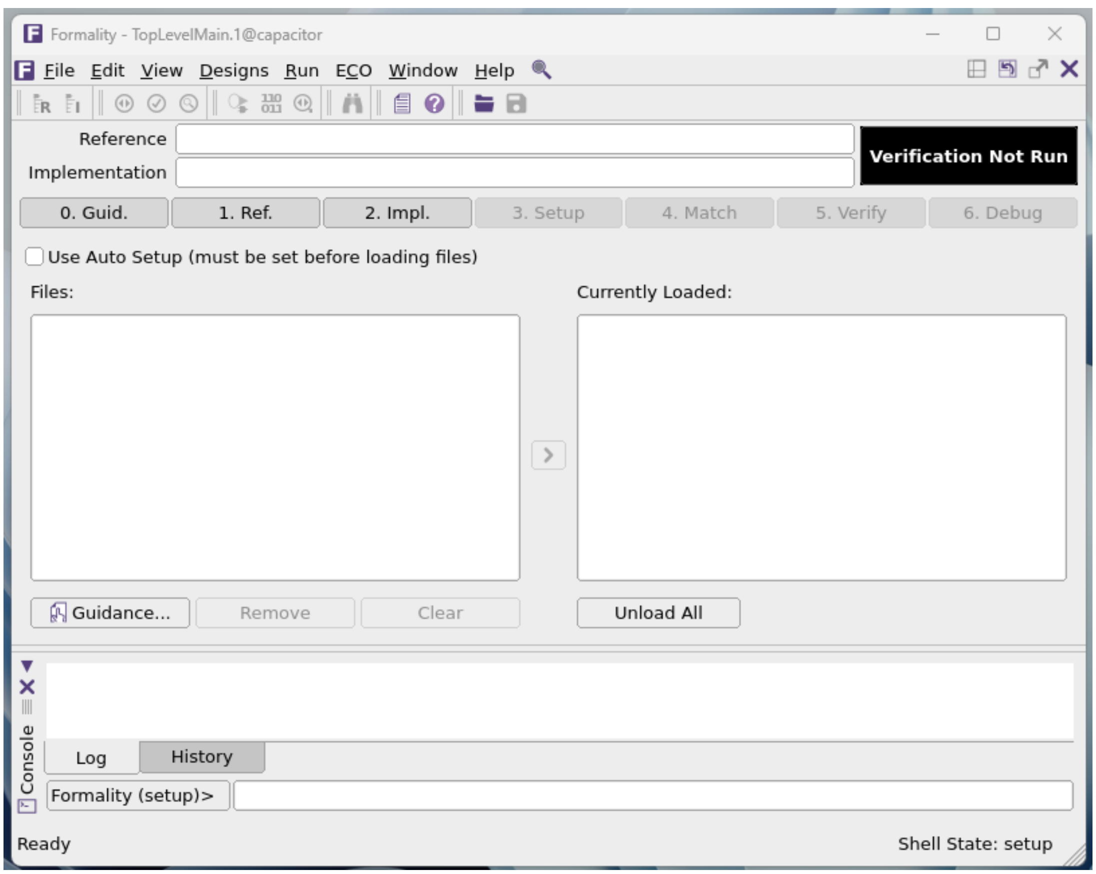


Fig.1. The Main Window of Formality GUI

Use the script, created to carry out complex commands. There is a script in **scripts** directory, which is located in **Post_syn** directory. The name of this script is **script.tcl,** which sources using the following command:

```
Formality(verify)> source ./script.tcl
```

Now understand Formality's general sections.

1. **Reference (Specify the Reference Design)**

Specifying the reference design involves reading in design files, optionally reading in technology libraries, and setting the top-level design.

The reference design is the design against which the transformed (implementation) design is compared. The reference design is the RTL source file named johnson.v file in the “Read Design Files” Tab. Click Verilog button, choose and Open this file (Fig 2). After that, choose the file in “Files Tab” and load it into the tool (Fig 3).

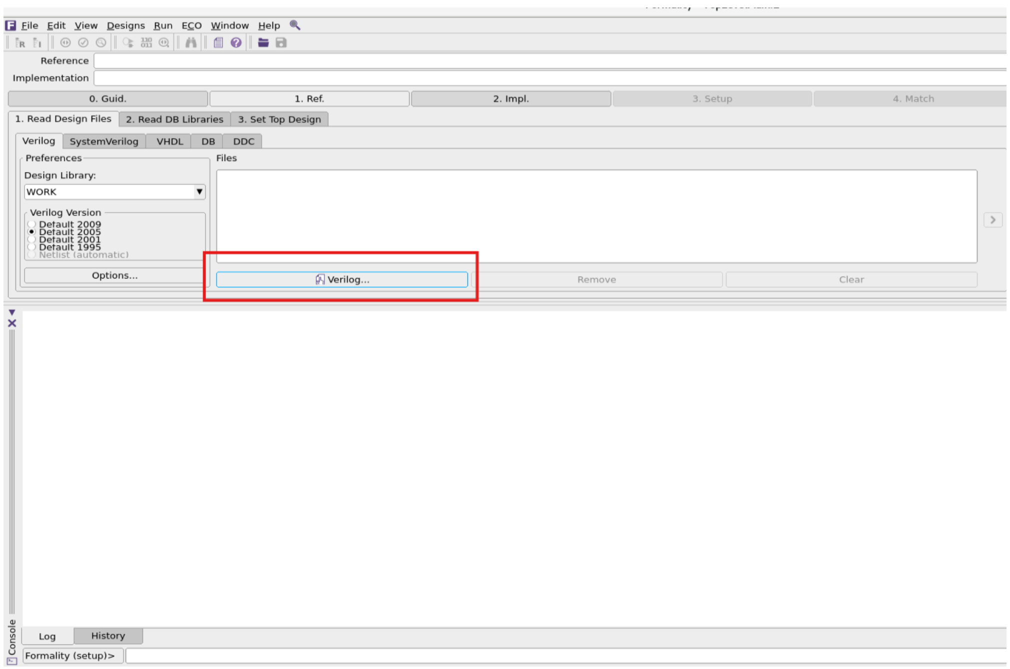


Fig.2. Set up Verilog Source in Reference.

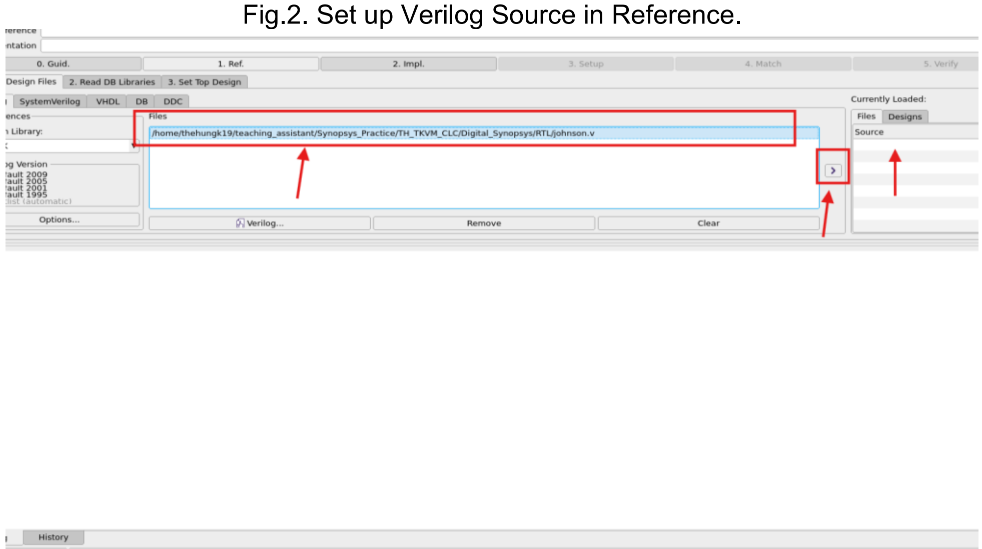


Fig.3. Load Verilog Source in Reference.

Click to Set Top Design of reference. Select Design Library in Section 1, choose a design johnson in Section 2 and click Set Top button (Fig 5). After setting Design Top, the green tick appearing means setting up Reference successfully.

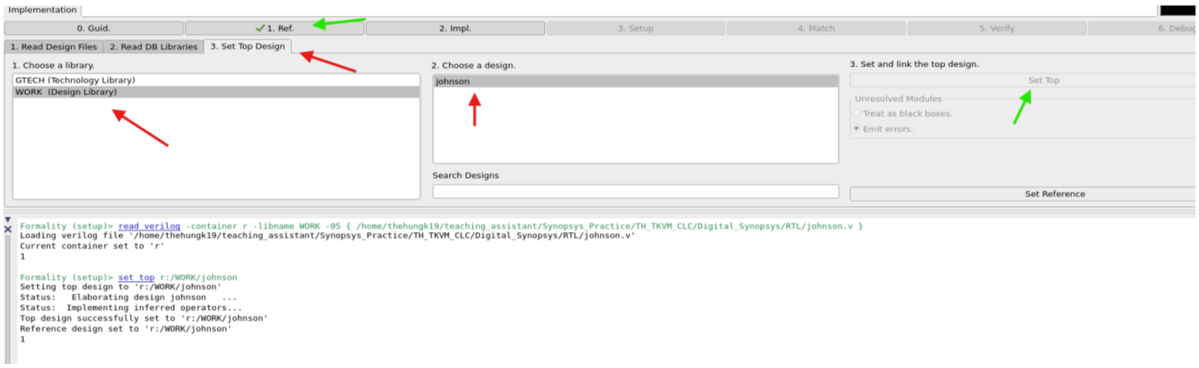


Fig.5. Set Top

2. Implementation (Specify the Implementation Design)

The procedure for specifying the implementation design is identical to that for specifying the reference design. The implementation design is the gate level netlist after Design Compiler named this source file johnson_compiled.v file. Change “2. Impl” Tab, Click Open and Load this file (Fig. 6).

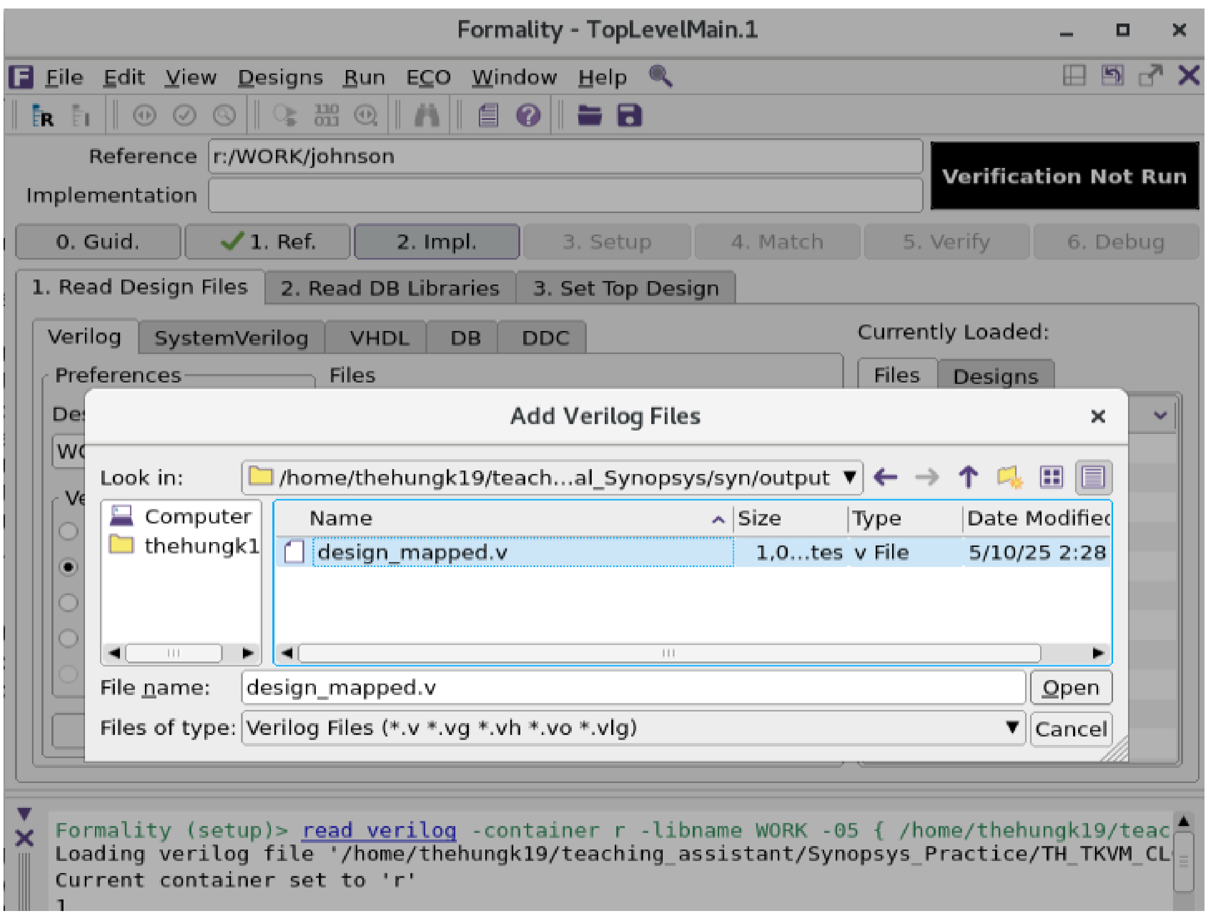


Fig.6. Set up gate level Verilog

Source the DB file. Select file **~/TH_TKVM_CLC/Digital_Synopsys/Lib/db/saed32rvt_tt1p05v25c.db**. Click Open and Load this file (Fig. 7).

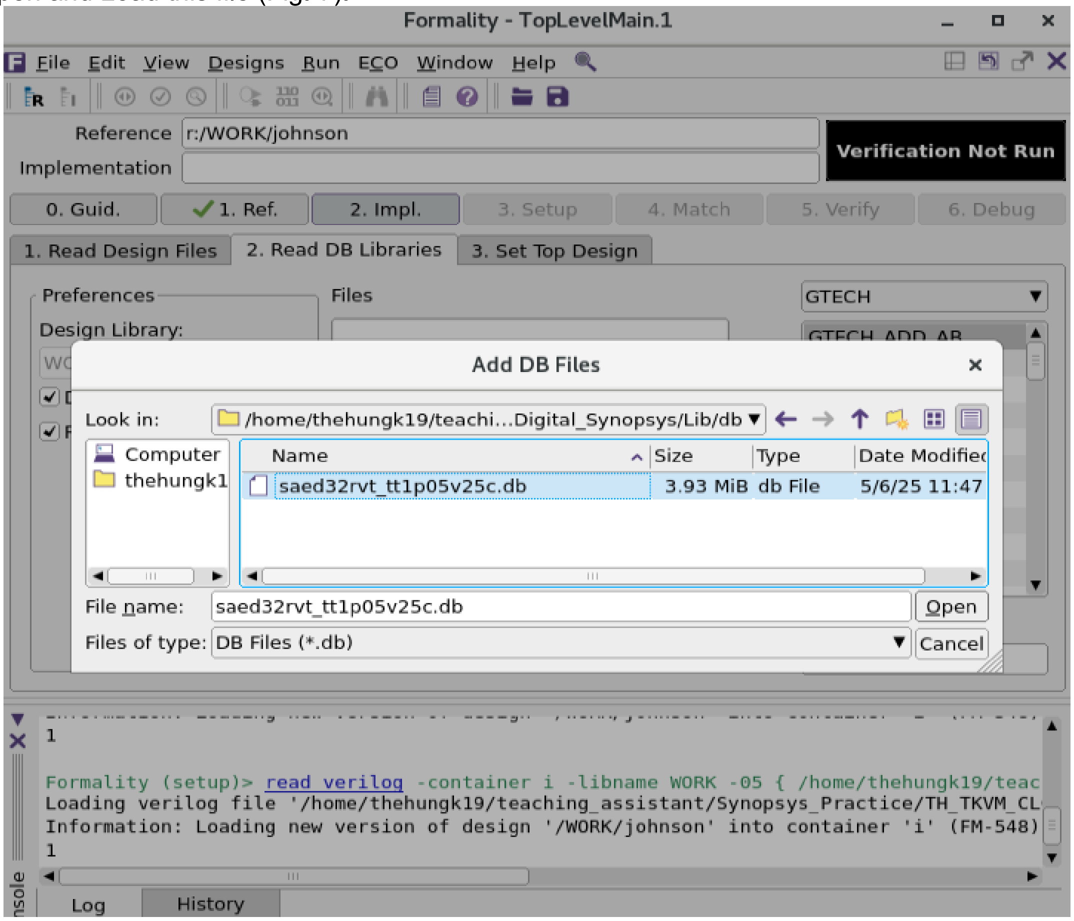


Fig.7. Set gate .db file

Set Top Design of implementation. Select Library WORK, choose a design johnson and Set and click Set Top button and Load this file.

3. Set up (Set Up the Design)

To Set up the design, click 3.set up and choose Set buttom (Fig.8 and Fig.9). Next, tick into Objects that you want to show and set “r” value to 0 in both of Reference and Implementation.

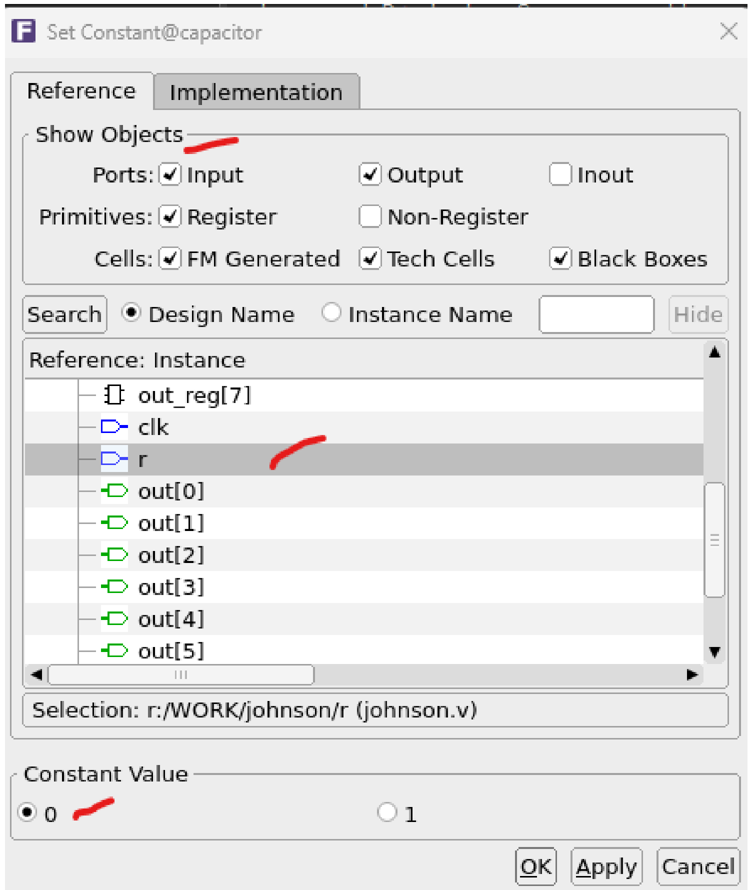


Fig.8. Set Up the Design window

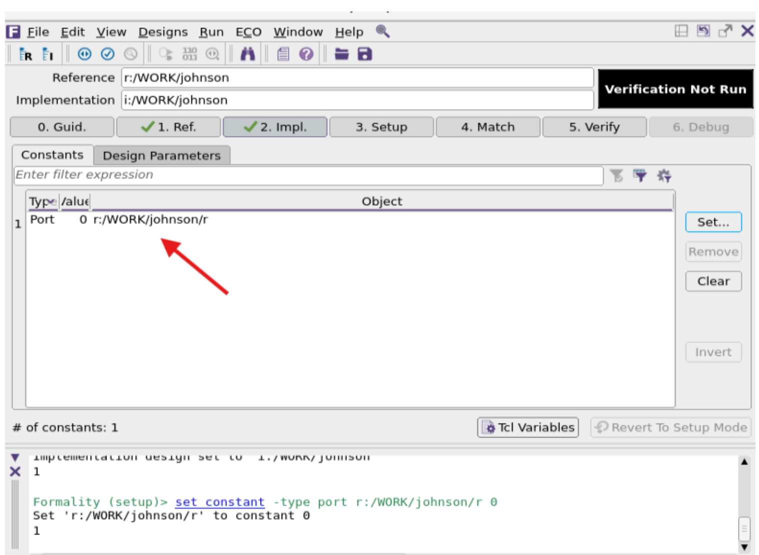


Fig.9. After Setting Up the Design.

4. Match (Match Compare Points)

Matching compare points is the process by which Formality segments the reference and implementation designs into logical units, called logic cones.

To match compare points between johnson.v and design_mapped.v, do the following:

In Main Window, click 4. Match and Run matching for match compare points.

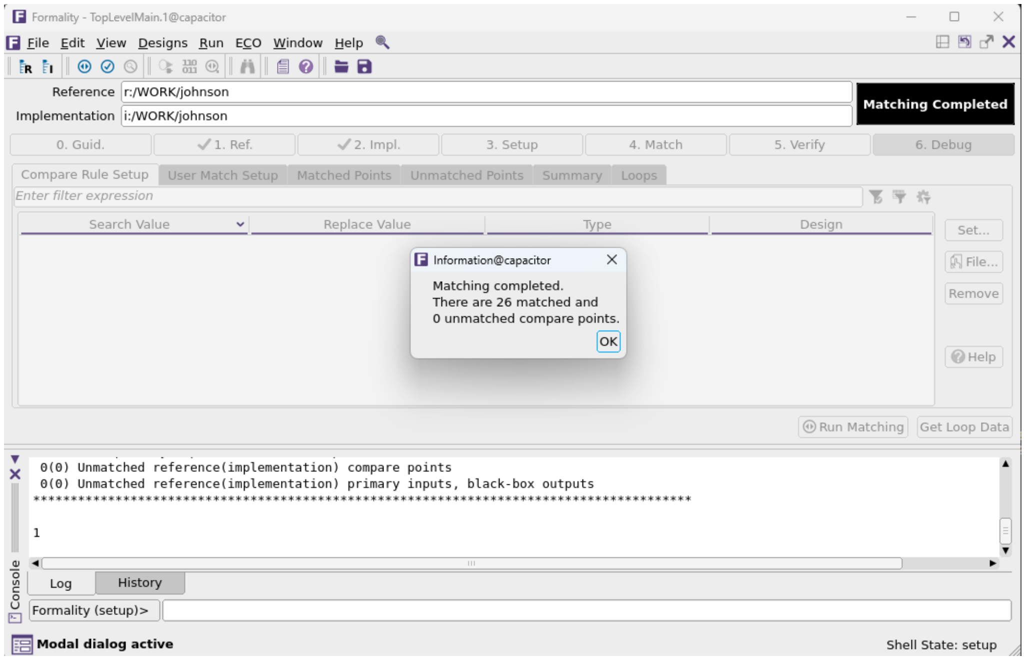


Fig.10. Match compare points

As seen from the massage shown in Fig.10, there are 0 unmatched compare points.

5. Verify (Verify the Designs)

When verify command is used, Formality attempts to prove design equivalence between an implementation design and a reference design. This section describes how to verify a design or a single compare point as well as how to perform traditional hierarchical verification.

On the main toolbar, click the Verify tab, then click Verify.

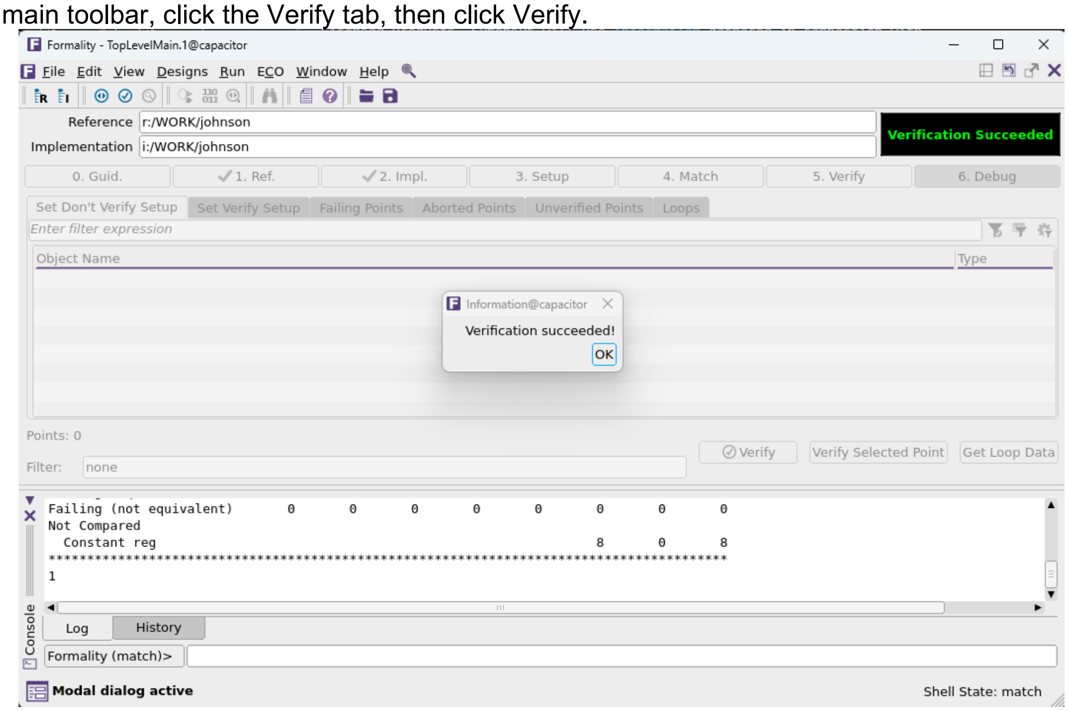


Fig.11. Verify the design

Verifying process completed successfully.

6. Debug (Optional)

During debugging, the exact points in the designs that exhibit the difference in functionality and then fix them must be found (Fig. 12).

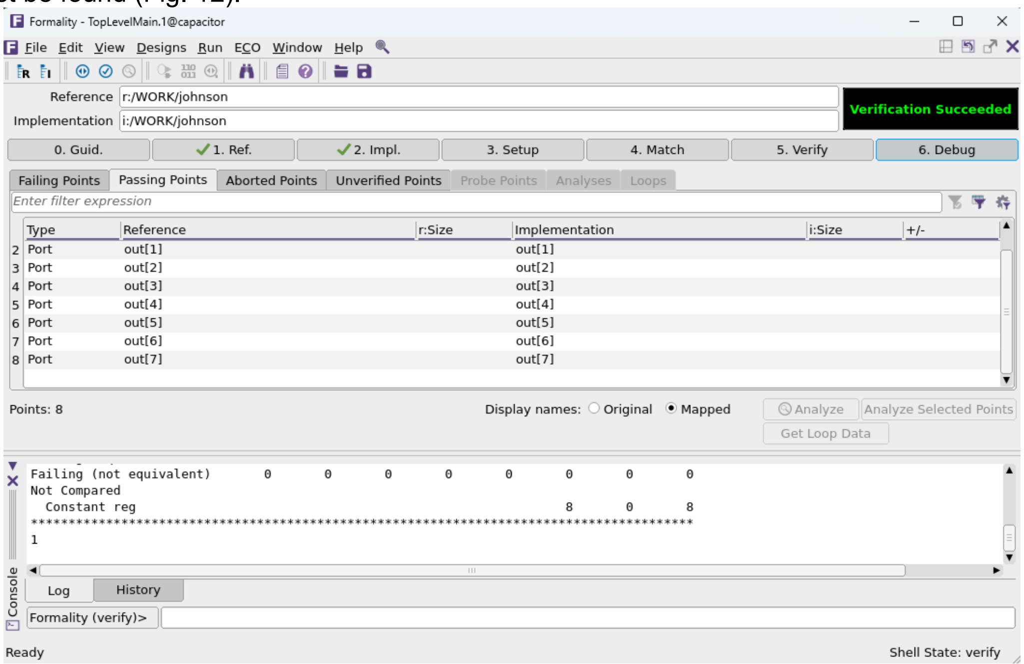


Fig.12. Debugging the design

Formality is able to simultaneously display reference and implementation Verilog views and mark differences and /or similarities (Fig. 13).

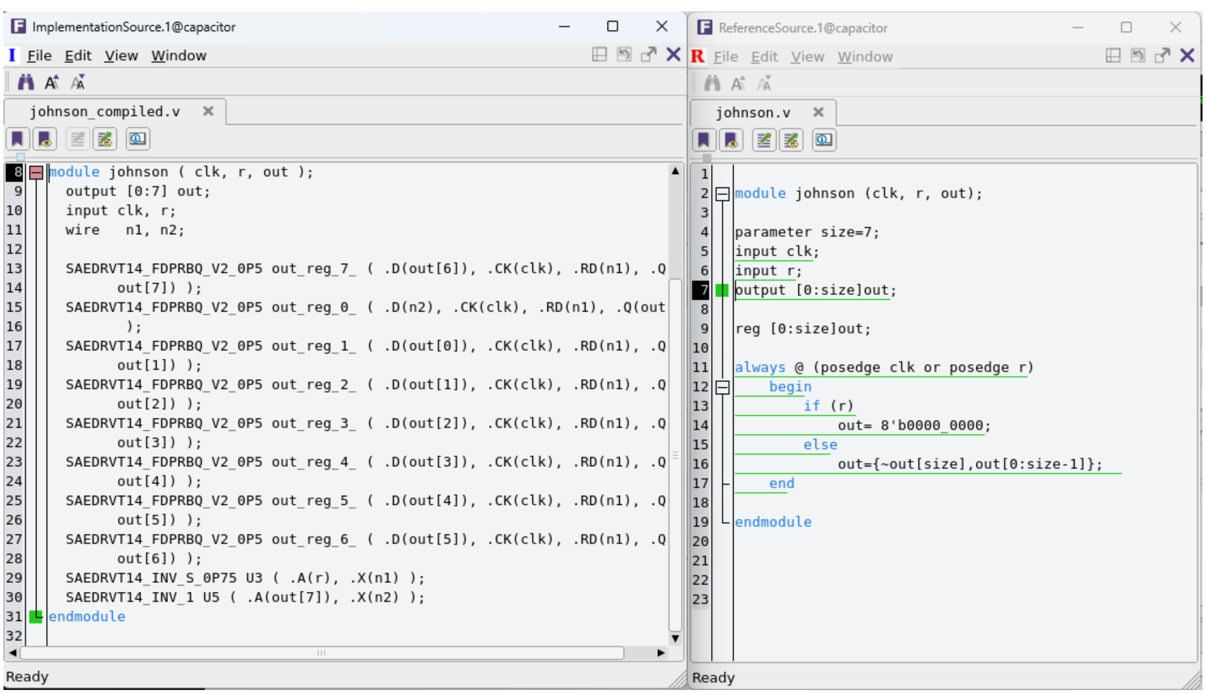


Fig.13(a). Implementation Verilog

Fig. 13(b). Reference Verilog

Besides, Formality can display schematic view and highlight reference object on it (Fig. 14).

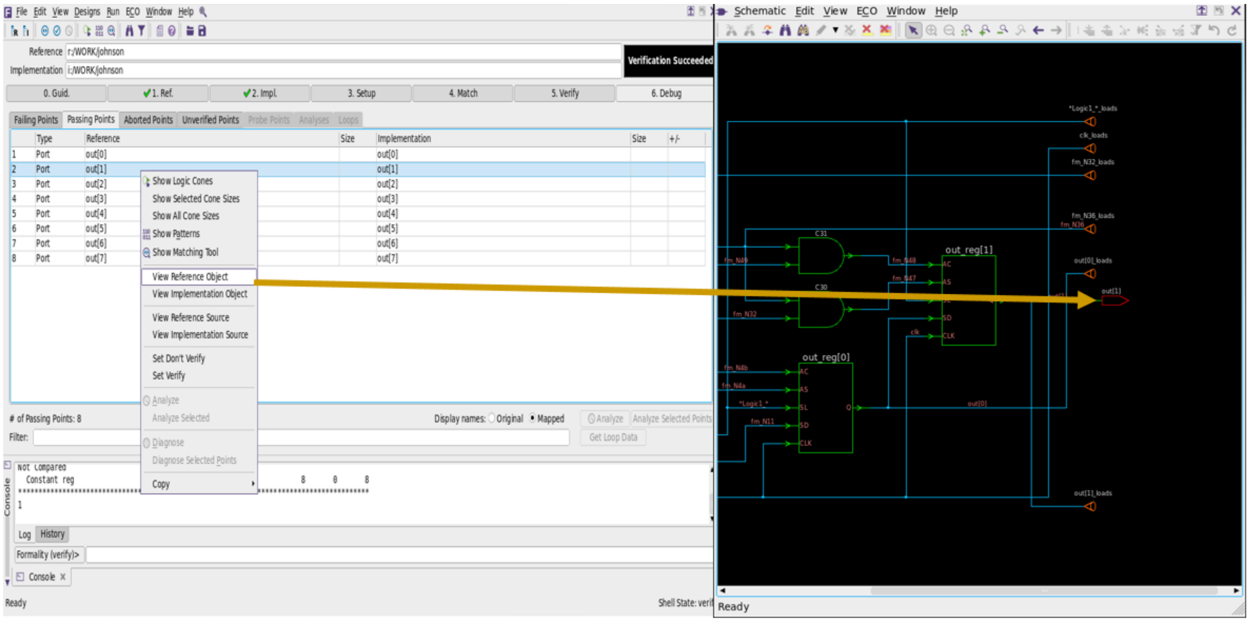


Fig.14. View Reference Object in schematic

To exit Formality, write exit in the command line.

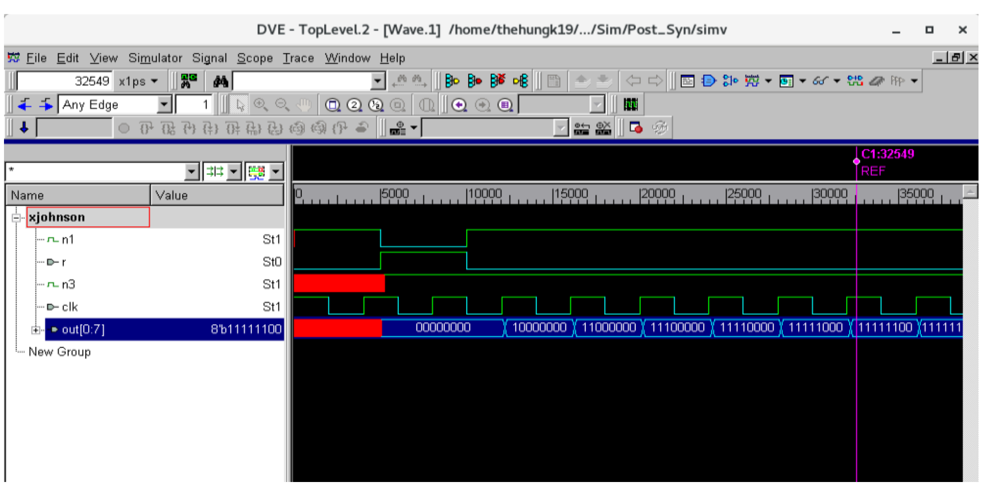
```
Formality (verify)> exit
```

7. Post-synthesis Simulation

In your current terminal, change your directory to **~/TH_TKVM_CLC/Digital_Synopsys/Sim/Post_Syn** and type the command line:

```
Bash>> vcs -gui -debug_all ../../RTL/johnson_tb.v ../../syn/output/design_mapped.v
../../Lib/verilog/saed32nm.v
Bash>> ./simv &
```

The results of the post-synthesis simulation appear with the waveforms.



Please check and compare with the waveforms of the RTL simulation step in Lab 1

B. Homework

Perform checking 2 designs after synthesizing them in Lab 2 by using Fomarlity and Post-synthesis simulation, including the combinational circuit and the 8-bit counter circuit.